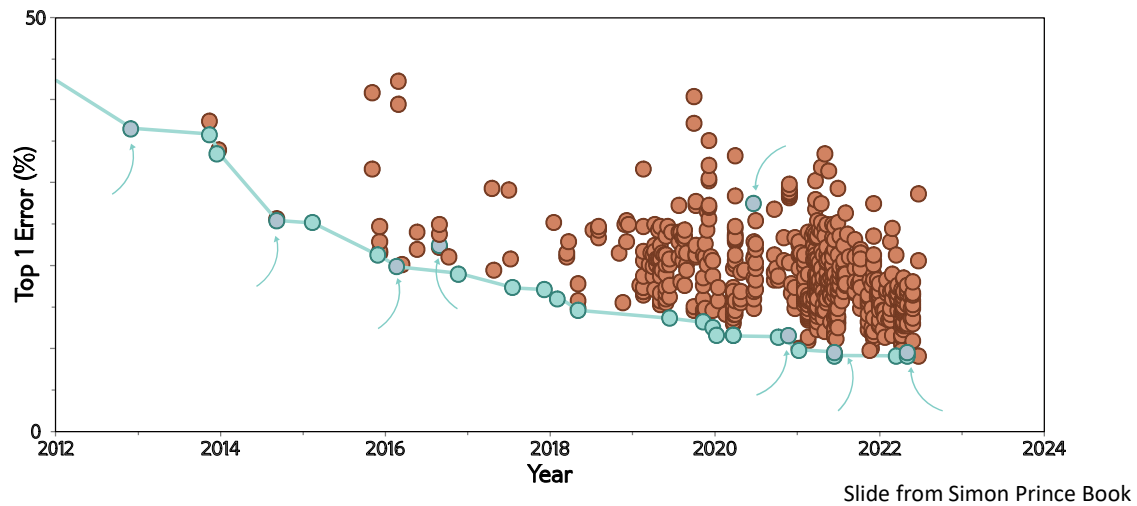


Image Classification Recent History



1

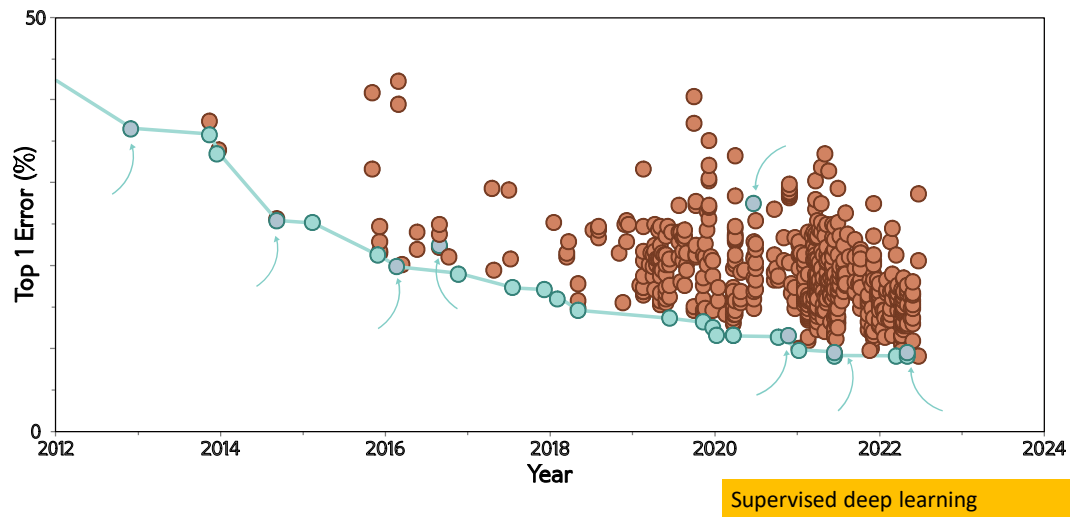
ImageNet Database



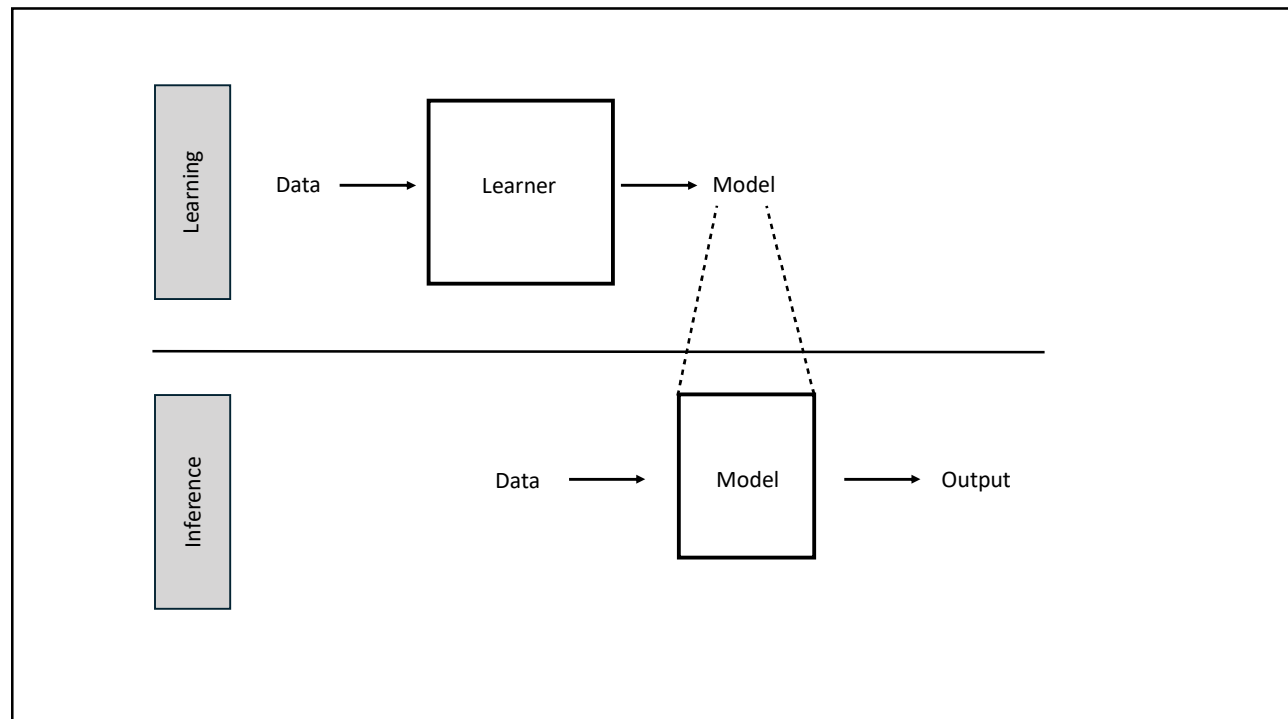
- 224×224 (size of images)
- 1,281,167 training images, 50,000 validation images, and 100,000 test images
- 1000 classes

2

Image Classification Recent History



3



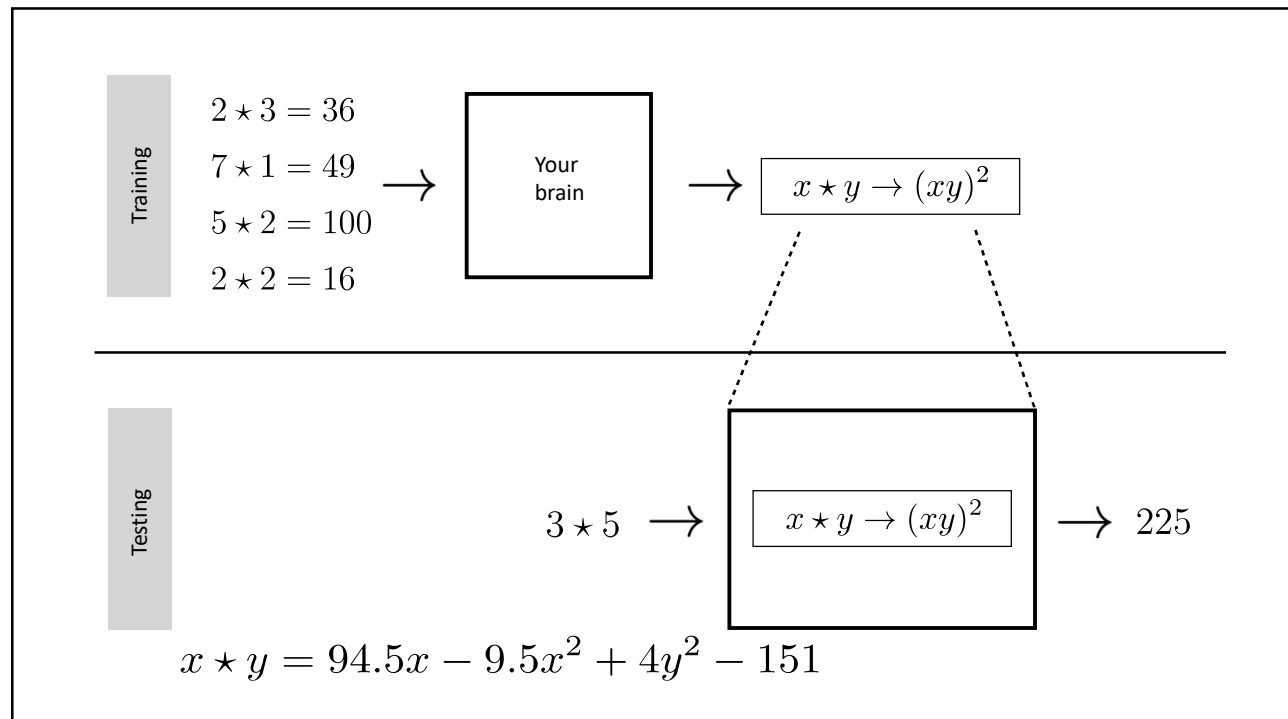
4

What does $\star$ do?

Hold your answer if you
have seen this before

$$\begin{aligned} 2 \star 3 &= 36 \\ 7 \star 1 &= 49 \\ 5 \star 2 &= 100 \\ 2 \star 2 &= 16 \end{aligned}$$

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Learning from examples

(aka supervised learning)

Training data

{input: [2, 3], output :36}
 {input: [7, 1], output :49}
 {input: [5, 2], output :100}
 {input: [2, 2], output :16}



Learner



f

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Learning from examples

(aka supervised learning)

Training data

$\{x^{(1)}, y^{(1)}\}$

$\{x^{(2)}, y^{(2)}\}$

$\{x^{(3)}, y^{(3)}\}$

...



Learner

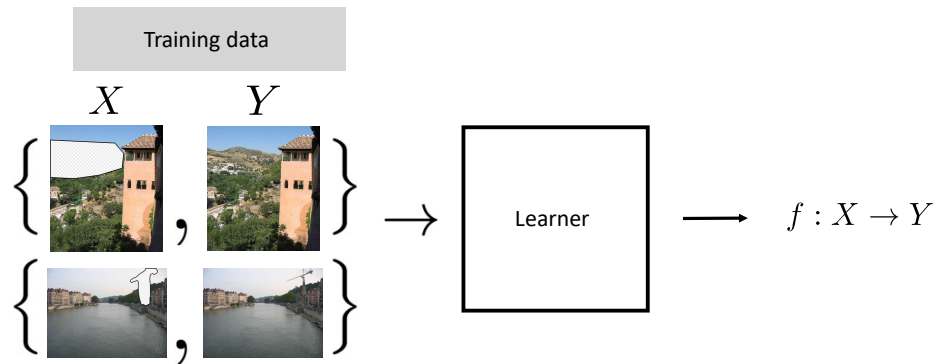


$f : X \rightarrow Y$

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Learning from examples

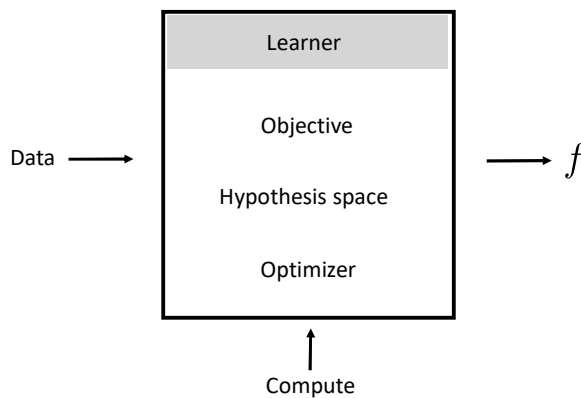
(aka supervised learning)



One can learn without examples (e.g. by specifying properties, or by reward functions)

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Key Ingredients



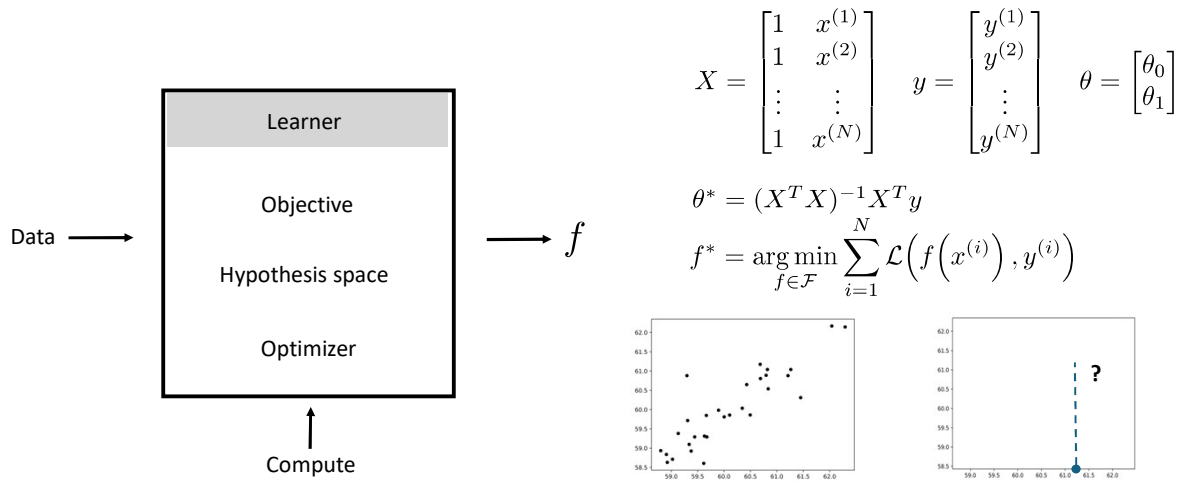
What does it mean for the learner to succeed?

What kind of space of possibilities we want to consider, and how do we parametrize them?

How do we search the space to get the desired objective?

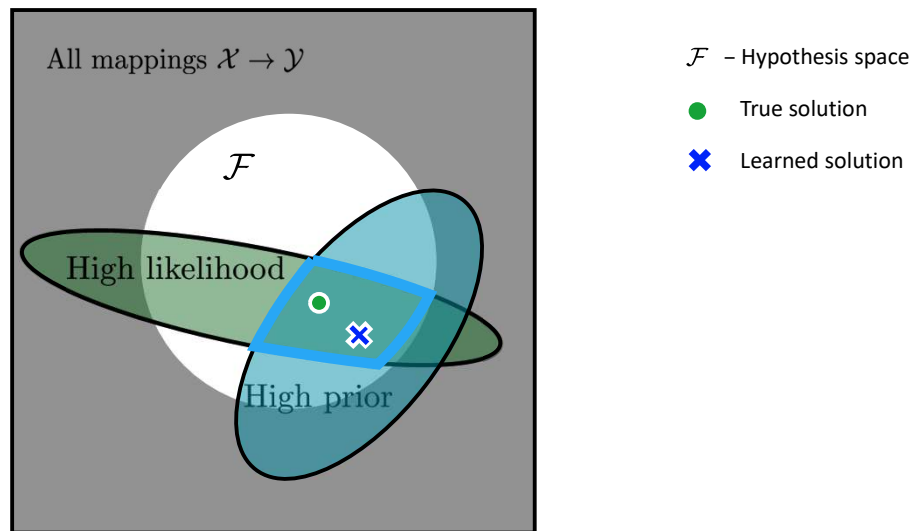
10

Example



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Data, Priors, and Hypotheses

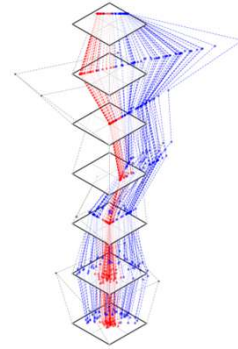


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Artificial Neural Nets

○ ANNs are a particular (simple) way of fixing the hypothesis, and a consistent way of optimizing

- They are universal approximators
- Can be trained using a simple algorithm (stochastic gradient descent)
- Run efficiently on parallel hardware.
- Inductive biases
- Build useful abstraction



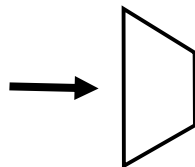
13

Learning for vision

How do you represent the input and output?

1. What is the objective?
2. What is the hypothesis space? (e.g., linear, polynomial, neural net?)
3. How do you optimize? (e.g., gradient descent, Newton's method?)
 - What's the loss function
4. What data do you train on?

$L[\phi]$ ← Returns a scalar that is smaller when model maps inputs to outputs better

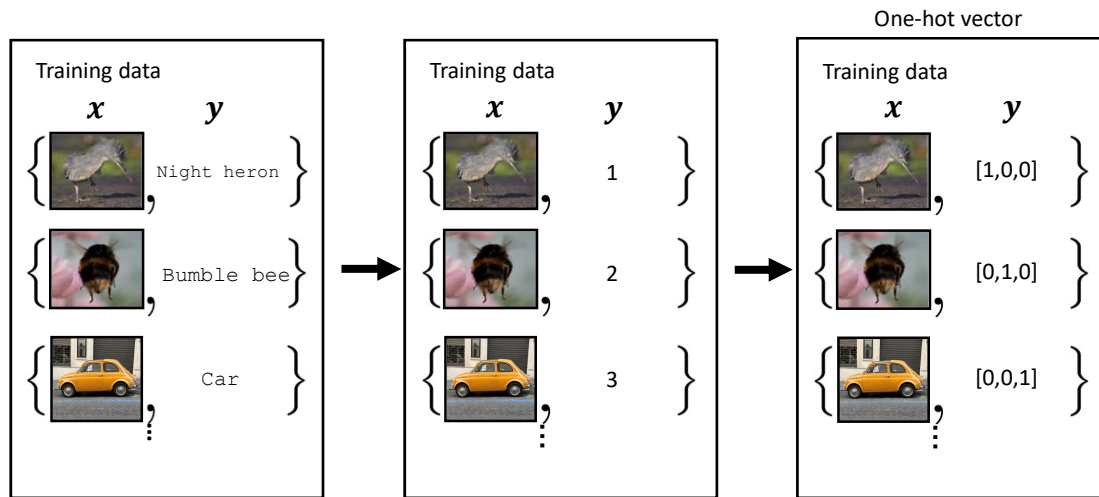


→ Guitarfish

$$\hat{\phi} = \underset{\phi}{\operatorname{argmin}} [L[\phi]]$$

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How to represent class labels?



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Loss functions

- ~~Model predicts output y given input x~~
- Model predicts a conditional probability distribution over outputs y given inputs x .
- Loss function aims to make the outputs have high probability

$$Pr(y|x)$$

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Recipe for loss functions

1. Choose a suitable probability distribution $Pr(y|\theta)$ that is defined over the domain of the predictions y and has distribution parameters.
2. Set the machine learning model $f[x, \phi]$ to predict one or more of these parameters so $\theta = f[x, \phi]$ and $Pr(y|\theta) = Pr(y|f[x, \phi])$.
3. To train the model, find the network parameters $\hat{\phi}$ that minimize the negative log-likelihood loss function over the training dataset pairs $\{x_i, y_i\}$:

$$\hat{\phi} = \underset{\phi}{\operatorname{argmin}} [L[\phi]] = \underset{\phi}{\operatorname{argmin}} \left[- \sum_{i=1}^I \log[Pr(y_i | f[x_i, \phi])] \right]$$
4. To perform inference for a new test example x , return either the full distribution $Pr(y|f[x, \hat{\phi}])$ or the maximum of this distribution

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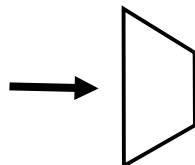
What should the loss be?

0-1 loss (number of misclassifications)

$$\mathcal{L}(\hat{y}, y) = \mathbb{1}(\hat{y} \neq y) \quad \leftarrow \text{discrete, NP-hard to optimize!}$$

Cross entropy

$$\mathcal{L}(\hat{y}, y) = H(y, \hat{y}) = - \sum_{k=1}^K y_k \log \hat{y}_k \quad \leftarrow \text{continuous, differentiable, convex}$$



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